

Topics

1) Hibernate

- Hibernate is a open-source object relational mapping framework.
- It allows java developer to interact with relational database using java objects, instead of sql queries.

2) API (Application programming Interface)

- Api is a set of rules and protocols that allows different software applications to communicate and exchange data with each other.

3) RestFul API

- Rest Api is a type of web API that follows (Representational state transfer) architecture.
- It allows two system to communicate over HTTP protocol using standard methoda like GET, POST, PUT, DELETE.
- GET - retrive data POST - create new data PUT - update existing data DELETE - delete data

4) JDBC

- Java Database Connectivity is an API that used to connect and interact with databases using sql queries.

5) SOLID Principles

Solid principles are five design principles in OOP that help developers to creates software that is easy to maintain , scalable and flexible.

1. Single responsibility principle Each function should have a single, well-defined responsibilty
2. Open-based principles A class should be open for extension but closed for modification
3. Liskov substitution principle Object of a superclass can be replaced with repect of its subclsss without affecting the programs architecture
4. Interface segregation principle keep interfaces small and focused
5. Dependency inversion control High-level modules should not depend on low-level modules both should depend on abstraction

6) Agile Methodology

- It is project management approach that breaks the project into phrases or sprints.
- It is flexible approach that focuses on continous collaboration, customer feedback.

7) Microservices

- It's a software development approach that breaks an application into independent services that communicate with each other.

8) Features of Java

- Java is a simple, object-oriented, and platform-independent programming language.
- It is secure, robust, portable, high-performing, and supports multithreading.
- Java programs can run on any device using the Java Virtual Machine (JVM).

9) JDK (Java Development Kit)

- JDK is a software development kit used to develop Java applications. It contains JRE, development tools like compilers (javac), debuggers, and other tools needed for Java programming.

10) JVM (Java Virtual Machine)

- JVM is an engine that runs Java bytecode. It provides a runtime environment, loads code, verifies code, executes code, and provides features like garbage collection and memory management.

11) JRE (Java Runtime Environment)

- JRE provides the environment to run Java applications. It contains JVM and essential libraries, but does not have development tools like the compiler.

12) Garbage Collection

- Garbage Collection in Java is the process of automatically identifying and removing unused or unreachable objects from memory to free up resources and improve performance.